

Risk Premium Inference from (Noisy) Option Price Panels

First version: June 16, 2018

Previous Version: August 6, 2026

This version: September 25, 2026

Brief Description of Research Idea

Compare the finite sample performance of implied state vs. particle filtering when conducting joint $\mathbb{P}$ - $\mathbb{Q}$ estimation on noisy option price panels focusing on the implications this has for recovered (equity) risk premium components and their error distributions/confidence bands.

Tentative Outline of Building Blocks

1. Implement an efficient simulation scheme for a Heston (Heston, 1993) state space simulation and associated option price panels.
2. Implement IS estimation à la Boswijk et al. (2016).
3. Implement particle filtering à la Hurn et al. (2015).
4. No noise/best case.
5. Noisy option panel case with an assorted menu of assumptions regarding option price error distributions.

Rationale

- A. Noisy option data is something I had to deal with both for the first two papers, outside academia and having spent some more time looking at post 2018 academic work it seems that this is an emerging topic with a few papers that touch on this in a different context: Andersen et al. (2021), Duarte et al. (2024).
- B. Topic itself ties in well with the other two papers in my thesis.
- C. I have sufficient background to work on this independently.

1 Assumed Data Generating Process And Option Pricing

We consider a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}\}_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions¹. Under the historical probability measure, $\mathbb{P}$, the dynamics of the log-price process $\log(S_t)$ and of the stochastic volatility process V_t , which make up the state vector $X_t = [\log(S_t), V_t]$, are assumed to be described by the pair of s.d.e.:

$$d\log(S_t) = \left((r_t - q_t) + (\eta - 1/2) V_t \right) dt + \sqrt{V_t} dW_{t,1}^{\mathbb{P}}; \quad (1.1)$$

$$dV_t = \kappa(\bar{V} - V_t) dt + \sigma_V \sqrt{V_t} \left(\rho dW_{t,1}^{\mathbb{P}} + \sqrt{1 - \rho^2} dW_{t,2}^{\mathbb{P}} \right). \quad (1.2)$$

$(dW_{t,1}^{\mathbb{P}}, dW_{t,2}^{\mathbb{P}})$ are $\mathcal{F}_t$ -adapted Brownian motion processes. $\{r\}_{t \geq 0}$ and $\{q\}_{t \geq 0}$ are $\mathcal{F}_t$ -adapted processes (e.g., deterministic) which describe the risk-free rate of interest and the continuous dividend yield respectively. η is a parameter which governs the size of the equity risk premium. The local stochastic volatility process mean reverts at a mean reversion rate κ to the long run mean level $\bar{V}$. σ_V governs the volatility of volatility and ρ represents the instantaneous correlation between returns and volatility.

Under the risk neutral probability measure, $\mathbb{Q}$, the model dynamics are assumed to be:

$$d\log(S_t) = \left((r_t - q_t) - 1/2 V_t \right) dt + \sqrt{V_t} dW_{t,1}^{\mathbb{Q}}; \quad (1.3)$$

$$dV_t = \kappa(\bar{V} - V_t) dt + \eta_V V_t dt + \sigma_V \sqrt{V_t} \left(\rho dW_{t,1}^{\mathbb{Q}} + \sqrt{1 - \rho^2} dW_{t,2}^{\mathbb{Q}} \right). \quad (1.4)$$

$(dW_{t,1}^{\mathbb{Q}}, dW_{t,2}^{\mathbb{Q}})$ are Brownian motions under the (equivalent) martingale pricing measure $\mathbb{Q}$.

Jointly, 1.3 and 1.4 are known as the Heston model, which features local stochastic volatility with a leverage effect while providing semi-closed form solutions for option prices. This model is a "staple" stochastic volatility model for equity index modeling. It is used in a large number of empirical studies, has a number of well documented limitations, some of which have been remedied by enriching the model specification through the addition of jumps and/or other (latent) stochastic states.

Given the focus of the study, attention is restricted to the simplistic set-up of the Heston model for a host of reasons. On the one hand, for most Heston model extensions, econometric approaches to parameter estimation which employ option price data come with increased computational costs. Since in the current set-up this model is known

¹See e.g., Protter (2005), p. 3.

to be the data generating process, there's no potential misspecification issue at play. Secondly, extensions come with specific estimation challenges which typically require tailored estimation designs to obtain well-behaved parameter estimates therefore limiting the comparability of results obtained throughout the study.

1.1 Parameter Sets

Let θ denote the vector of parameters featured in the model. We assume $\theta \in \Theta$ where Θ is a compact set from the domain of valid data-generating process parameters. For the Heston model specification previously introduced, the value of θ corresponding to the data-generating process introduced in (1.1) - (1.4) is $\theta = [\eta, \kappa, \bar{V}, \sigma_V, \rho, \eta_v]$. Note that subsets of the parameter set play a role in either the $\mathbb{P}$ -measure specification, the $\mathbb{Q}$ -measure specification or in both. E.g., the Heston parameter set can be decomposed into 3 disjoint subsets: $\theta = [\theta^{\mathbb{P}}, \theta^{\mathbb{P} \cap \mathbb{Q}}, \theta^{\mathbb{Q}}]$, where $\theta^{\mathbb{P}} = \eta$, $\theta^{\mathbb{P} \cap \mathbb{Q}} = [\kappa, \bar{V}, \sigma_V, \rho]$, and $\theta^{\mathbb{Q}} = \eta_v$.

1.2 State Vector

Assuming we have a time series of state vector observations:² $X_1, \dots, X_T$, observed at equally-spaced timepoints $t_1, \dots, t_T$ with $t_{i+1} - t_i = \Delta > 0, \forall i \geq 1$. Denote by $f(X_{i+1}|X_i; \theta^{\mathbb{P}}, \theta^{\mathbb{P} \cap \mathbb{Q}})$ the transition density of the state vector X_t under the historical probability measure. Note that the transition density depends on $\theta^{\mathbb{P}}$ and $\theta^{\mathbb{P} \cap \mathbb{Q}}$ which represent the subset of parameters that describe the model dynamics under $\mathbb{P}$.

By using option price data the full set of parameters – including the additional ones that characterize model dynamics under the risk neutral pricing measure – can be estimated, i.e., inference on $\theta^{\mathbb{Q}}$ can be conducted while at the same time increasing sample identification potential for the $\theta^{\mathbb{P} \cap \mathbb{Q}}$ parameter set.

1.3 Option prices

Suppose that, in addition to time series observations of returns, synchronous observations of option price panel data and time series data on the underlying are available. Denote the option prices by $p_i(T, K)$ where T, K represent the maturity and the strike of the option contract. Option prices are commonly quoted in terms of their Black-Scholes implied volatility. Let $P(X_i, T, K; \theta^{\mathbb{P} \cap \mathbb{Q}}, \theta^{\mathbb{Q}})$ be the corresponding model price, e.g., the option price generated by the base model specification under the equivalent martingale

²In practice such a series never exists as the volatility, V_t , is an unobservable, i.e., latent state process

pricing measure for given state vector instance X_i and parameter set values $\theta^{\mathbb{P} \cap \mathbb{Q}}$ and $\theta^{\mathbb{Q}}$. If option prices were observed without error and the model is not misspecified, the observed option prices would coincide with their model-based counterparts. In practice the two prices differ. Typical modelling approaches assume away model misspecification and attribute the difference to measurement errors (e.g., microstructure noise). This assumption can in practice have consequences for the sample properties of the estimator of choice and thus warrants further consideration.

1.3.1 Option Prices with Observation Errors

Suppose that, at each observation date t_i , a panel of option price quotes is observed for a set of contracts from ranging from $j = 1$ to $j = q_i$ where $q_i \in \mathbb{N}^*$. This set is denoted by $\mathcal{C}_i = \{(T_i^{(j)}, K_i^{(j)})\}_{j=1}^{q_i}$ and the resulting price vector by $p_i^{\text{obs}} = \left(p_i^{(1),\text{obs}}, \dots, p_i^{(q_i),\text{obs}}\right)^\top$, where $p_i^{(j),\text{obs}} = p_i^{\text{obs}}(T_i^{(j)}, K_i^{(j)})$ is the observed price for option contract j .

Similarly, let $p_i^0 = p_i^{\text{model}} = \left(p_i^{(1),0}, \dots, p_i^{(q_i),0}\right)^\top$ be the vector of corresponding model-based prices implied by $P(X_i, T, K; \theta^{\mathbb{P} \cap \mathbb{Q}}, \theta^{\mathbb{Q}})$. Let $\sigma_i^0 = \left(\sigma_i^{(1),0}, \dots, \sigma_i^{(q_i),0}\right)^\top$ denote the corresponding Black-Scholes implied volatility vector.

The difference between p_i^{obs} and p_i^{model} is interpreted as option observation error. This difference can be, inter alia, attributed to microstructure effects or stale quotes. Duarte et al. (2024) suggests that such frictions can be quantitatively important for inference involving option-implied risk premia.

The proposed simulation design therefore does not commit ex-ante to a single parametric error law but instead considers four observed-panel specifications generated from the same clean model-implied panel. Each specification is meant to reflect empirical concerns raised by the recent literature while keeping the data generating exercise tractable. The four specifications are introduced in implied-volatility space. The corresponding noisy prices are then obtained through the Black-Scholes pricing formula and projected back into the no-arbitrage price interval. This keeps the treatment of all four observation error specifications internally consistent while preserving both price and implied-volatility representations of the noisy option panel.

1.3.1.1 Design A: Low I.I.D. noise

First, a low-noise benchmark is introduced in which observation errors are independent across contracts, but the marginal scale of the error is allowed to vary with moneyness

and maturity. This reflects the fact that option observation errors need not have the same relative impact across the surface and can enter the behaviour of option-based estimators through local features of the panel, in particular for contracts close to the spot or forward price as discussed in Chong and Todorov (2024). For $j = 1, \dots, q_i$, let

$$m_i^{(j)} = \log \left(\frac{K_i^{(j)}}{F_i^{(j)}} \right), \quad \mathcal{T}_i^{(j)} = T_i^{(j)} - t_i,$$

where $F_i^{(j)}$ denotes the corresponding forward price. The marginal implied-volatility observation error scale is specified as

$$s_{\sigma,i}^{(j)} = s_{\sigma}(m_i^{(j)}, \mathcal{T}_i^{(j)}) = \alpha_0 \left(1 + \alpha_m |m_i^{(j)}| + \frac{\alpha_{\mathcal{T}}}{\sqrt{\max(\mathcal{T}_i^{(j)}, \mathcal{T}_{\min})}} \right). \quad (1.5)$$

The first specification is intended to mimic a liquid-market benchmark in which implied-volatility observation errors are independent across contracts conditional on the clean panel:

$$\tilde{\sigma}_i^{(j),A} = \max \left\{ \sigma_{\min}, \sigma_i^{(j),0} + s_{\sigma,i}^{(j)} Z_i^{(j)} \right\}, \quad Z_i^{(j)} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1). \quad (1.6)$$

In other words, this specification represents small quote observation errors whose scale varies with moneyness and maturity, without imposing dependence across contracts. It serves as the low-noise benchmark case.

The realised implications of the observation-error specifications are illustrated in Appendix B. In the low i.i.d. design, the perturbations are contract-specific and preserve the broad shape of the clean implied-volatility panel, while allowing the local noise scale to vary with moneyness and maturity.

1.3.1.2 Design B: Cross-sectionally correlated noise

A second observation error design allows for cross-sectional dependence in a given option panel for timepoint t_i by correlating errors across nearby strikes and tenors. This is largely motivated by the evidence and testing framework in Andersen et al. (2021), which explicitly considers spatial dependence in option observation errors. In this specification, the information content of a panel with q_i option prices is essentially not equivalent to that of q_i independent observations.

Let $\varepsilon_i^B = (\varepsilon_i^{(1),B}, \dots, \varepsilon_i^{(q_i),B})^\top$ and set

$$\tilde{\sigma}_i^{(j),B} = \max \left\{ \sigma_{\min}, \sigma_i^{(j),0} + \varepsilon_i^{(j),B} \right\}, \quad j = 1, \dots, q_i, \quad (1.7)$$

where

$$\varepsilon_i^B \sim \mathcal{N}(0, D_i R_i D_i), \quad (1.8)$$

with

$$D_i = \text{diag} \left(s_{\sigma,i}^{(1)}, \dots, s_{\sigma,i}^{(q_i)} \right).$$

The correlation matrix R_i is given by

$$R_{i,jk} = \exp \left(-\frac{|m_i^{(j)} - m_i^{(k)}|}{\ell_m} - \frac{|\mathcal{T}_i^{(j)} - \mathcal{T}_i^{(k)}|}{\ell_{\mathcal{T}}} \right). \quad (1.9)$$

This specification captures the possibility that nearby strikes and maturities share common quote or surface construction errors.

The cross-sectionally correlated design creates locally coherent distortions across nearby moneyness and maturity points. Appendix B reports the corresponding noisy option panels, signed perturbations and cross-sectional correlation structure.

1.3.1.3 Design C: Temporally persistent noise

The third observation error design allows for a persistent common component in the observation error, affecting the level, moneyness slope and maturity slope of the noisy implied volatility panel. This is consistent with the practical treatment of option panels as surfaces and with the broader evidence that option price noise can be quantitatively relevant for inference on risk premia as in Duarte et al. (2024).

Let $f_i = (f_{0,i}, f_{1,i}, f_{2,i})^\top$ evolve according to

$$f_i = A f_{i-1} + \xi_i, \quad \xi_i \sim \mathcal{N}(0, Q), \quad (1.10)$$

where A is diagonal with entries inside the unit circle. For contract j , define

$$b_i^{(j)} = \left(1, m_i^{(j)}, \mathcal{T}_i^{(j)} \right)^\top.$$

The raw implied-volatility observation error is

$$\tilde{\sigma}_i^{(j),C} = \max \left\{ \sigma_{\min}, \sigma_i^{(j),0} + \left(b_i^{(j)} \right)^\top f_i + u_i^{(j)} \right\}, \quad u_i^{(j)} \sim \mathcal{N} \left(0, \left(s_{\sigma,u,i}^{(j)} \right)^2 \right), \quad (1.11)$$

where $s_{\sigma,u,i}^{(j)}$ is allowed to depend on $m_i^{(j)}$ and $\mathcal{T}_i^{(j)}$.

The matrix A is the persistence matrix of the common observation-error factors. In the present specification it is taken to be diagonal, so that each component of f_i follows a separate first-order autoregressive process. The restriction that the diagonal entries of A lie inside the unit circle ensures stationarity. The matrix Q is the covariance matrix of the innovations ξ_i and controls the magnitude and possible co-movement of new shocks to the common components of the observation error.

The factor representation is a parsimonious way of introducing temporal dependence in the option panel observation errors. The vector f_i describes the common observation-error state at date t_i , while the loading vector $b_i^{(j)}$ determines how this common component affects option contract j , i.e., $\left(b_i^{(j)} \right)^\top f_i = f_{0,i} + m_i^{(j)} f_{1,i} + \mathcal{T}_i^{(j)} f_{2,i}$. The first component acts as a common level error, the second component as a moneyness-slope error, and the third component as a maturity-slope error. This type of low-dimensional factor representation is consistent with the empirical treatment of implied-volatility panels as objects whose movements can be summarised by a small number of common components; see, for example, Cont and da Fonseca (2002) and related PCA-based evidence for implied-volatility surface such as Avellaneda et al. (2020).

Compared to Designs A and B, the persistent-factor design C induces serially dependent quote distortions through (several) common component(s). Appendix B reports the realised factor paths and their effect on representative option contracts.

1.3.1.4 Design D: Variance dependent noise

The fourth observation error design extends Design C by allowing part of the temporally persistent common observation error to depend directly on the latent variance state. The purpose of this specification is to examine whether the implied-state estimator is more sensitive to observation errors which move systematically with the state being recovered from the option panel, while keeping the unconditional magnitude of the observation error comparable to that of Designs A–C.

Let

$$z_{V,i} = \frac{V_i - \bar{V}}{s_V}, \quad s_V^2 = \frac{\bar{V} \sigma_V^2}{2\kappa},$$

denote the standardised latent variance state under the stationary distribution of the variance process. Design D adds to the factor specification of Design C a fourth, variance linked common component,

$$f_{V,i} = \gamma_V z_{V,i},$$

where γ_V determines the magnitude of the dependence between the option observation error and the latent variance state.

The noisy implied volatility is then generated as

$$\tilde{\sigma}_i^{(j),D} = \max \left\{ \sigma_{\min}, \sigma_i^{(j),0} + \left(b_i^{(j)} \right)^\top f_i + \gamma_V z_{V,i} + u_i^{(j),D} \right\}. \quad (1.12)$$

Thus, Designs C and D share the same persistent level, moneyness-slope and maturity-slope factors, while Design D additionally allows a common component of the observation error to vary with the contemporaneous latent variance state. The residual idiosyncratic scale is adjusted so that the unconditional pointwise implied-volatility observation error remains comparable to that of the other designs. Design D therefore changes the dependence structure of the observation error without increasing its overall magnitude.

1.3.2 Implied-State C-GMM Estimation

The first estimation approach considered in the Monte Carlo exercise is an implied-state continuum GMM estimator in the spirit of Boswijk et al. (2016). The estimator uses the option panel at each observation date to recover a parameter-dependent proxy for the latent variance state and subsequently imposes moment restrictions based on the conditional characteristic function of the resulting state vector. This provides a natural benchmark for the particle-filter approach introduced below. The implied-state estimator summarises each option panel by a latent state implied from the cross-section of option prices, whereas the particle filter treats the option panel as part of the measurement system used to update a distribution of latent states.

Let

$$\theta_{\mathbb{Q}} := \left[\theta^{\mathbb{P} \cap \mathbb{Q}}, \theta^{\mathbb{Q}} \right]$$

denote the subset of parameters which enters the risk-neutral option pricing problem.

In the Heston specification considered, this corresponds to

$$\theta_{\mathbb{Q}} = [\kappa, \bar{V}, \sigma_V, \rho, \eta_V].$$

Given a candidate value of $\theta_{\mathbb{Q}}$, the observed option panel at date t_i can be used to construct an implied variance state. Let

$$\sigma_i^{\text{obs}} = \left(\sigma_i^{(1),\text{obs}}, \dots, \sigma_i^{(q_i),\text{obs}} \right)^{\top}$$

denote the vector of observed Black-Scholes implied volatilities corresponding to the contracts in $\mathcal{C}_i$. For a candidate variance level v , let $\sigma^{(j),\text{model}}(\log(S_i), v; \theta_{\mathbb{Q}})$ denote the model-implied Black-Scholes volatility of contract j obtained from the Heston pricing model. The implied variance state is defined as

$$\hat{V}_i^{\theta_{\mathbb{Q}}} = \arg \min_{v \in \mathcal{V}} \sum_{j=1}^{q_i} \left[\sigma_i^{(j),\text{obs}} - \sigma^{(j),\text{model}}(\log(S_i), v; \theta_{\mathbb{Q}}) \right]^2, \quad (1.13)$$

where $\mathcal{V}$ is the admissible domain for the variance state. In the absence of observation error, and evaluated at the true parameter vector, this inversion recovers the latent variance sequence up to numerical approximation error. In noisy option panels, $\hat{V}_i^{\theta_{\mathbb{Q}}}$ is instead a parameter-dependent transformation of the noisy option cross-section. This is the main channel through which option observation errors affect the implied-state estimator.

The option pricing step uses a maturity-specific fixed effective COS interval. The same interval is used when generating the option panels and when reconstructing the implied variance states, while the number of COS terms may differ between these two steps if this is supported by the numerical validation. The calibration of the fixed intervals and the numerical implementation of the state inversion are reported in Appendix D.

Define the observed log-return increment

$$R_i := \log(S_i) - \log(S_{i-1})$$

and collect the observable return and implied variance into the parameter-dependent estimation state:

$$X_i^{\theta_{\mathbb{Q}}} = \left(R_i, \hat{V}_i^{\theta_{\mathbb{Q}}} \right)^{\top}.$$

Although the state-implying step depends only on $\theta_{\mathbb{Q}}$, the transition law of $X_i^{\theta_{\mathbb{Q}}}$ under

the historical probability measure depends on the full parameter vector θ . Let

$$\varphi\left(s, X_i^{\theta_{\mathbb{Q}}}, \Delta; \theta\right) = \mathbb{E}_{\theta} \left[\exp\left(is^{\top} X_{i+1}^{\theta_{\mathbb{Q}}}\right) \mid X_i^{\theta_{\mathbb{Q}}}\right]$$

denote the conditional characteristic function of the next estimation state, where $s \in \mathbb{R}^2$. For the Heston model this conditional characteristic function is available in affine form under $\mathbb{P}$.

The implied-state C-GMM estimator is based on the martingale difference

$$e_i(s; \theta) = \exp\left(is^{\top} X_{i+1}^{\theta_{\mathbb{Q}}}\right) - \varphi\left(s, X_i^{\theta_{\mathbb{Q}}}, \Delta; \theta\right).$$

Following the characteristic-function approach to GMM estimation, this residual is multiplied by a continuum of instruments of the form

$$\exp\left(iu^{\top} X_i^{\theta_{\mathbb{Q}}}\right), \quad u \in \mathbb{R}^2.$$

Let $\tau = (u, s)$ denote the combined argument of the continuum of moment conditions. The moment function is

$$h_i(\tau; \theta) = \exp\left(iu^{\top} X_i^{\theta_{\mathbb{Q}}}\right) \left[\exp\left(is^{\top} X_{i+1}^{\theta_{\mathbb{Q}}}\right) - \varphi\left(s, X_i^{\theta_{\mathbb{Q}}}, \Delta; \theta\right) \right].$$

At the true parameter value, and under the maintained model assumptions, these moments satisfy

$$\mathbb{E} \left[h_i(\tau; \theta_0) \right] = 0, \quad \text{for all } \tau.$$

Let

$$\widehat{h}_T(\tau; \theta) = \frac{1}{T-1} \sum_{i=1}^{T-1} h_i(\tau; \theta)$$

denote the corresponding sample moment. The first-step implied-state C-GMM criterion is

$$Q_T^{(1)}(\theta) = \int \left| \widehat{h}_T(\tau; \theta) \right|^2 \pi(\tau), d\tau,$$

where π is an integrating density over the continuum of moments. The first-step estimator is then

$$\widehat{\theta}_{\text{IS}} = \arg \min_{\theta \in \Theta} Q_T^{(1)}(\theta). \tag{1.14}$$

The numerical evaluation of (1.14), including the construction of the implied variance

states, the conditional characteristic function and the integration rule for the continuum of moments, is described in Appendix D.

The distinction between the clean and noisy option panels is particularly relevant for this estimator. In Design A, independent contract-level observation errors perturb the implied variance state locally through the cross-sectional inversion in (1.13). In Design B, cross-sectional dependence in the observation errors can generate coherent distortions across nearby moneyness and maturity points, which may be partially absorbed as changes in the implied state. In Design C, persistent common factors in the option observation error may generate serially dependent distortions in the recovered implied variance sequence. Design D additionally allows the common level component of the observation error to depend on the latent variance state itself, providing a stress test in which part of the measurement error can move systematically with the state being recovered. The Monte Carlo comparison therefore evaluates sensitivity not only to the magnitude of option observation error, but also to its cross-sectional, temporal and state-dependent structure.

1.3.3 Particle Filter Based Estimation

Estimation contexts which make use of option prices (that are nonlinear, often semi-analytic expressions featuring model parameters and partially observable state vectors) can be computationally demanding. The models used to describe index returns for option pricing purposes typically feature latent stochastic state processes which require filtering or implying. This increases the computational cost of econometric estimation exercises as one has to devise parsimonious estimators which can work with non-analytic expressions of state vector observations by relying on stable and robust numerical procedures.

Two other hurdles are typically encountered when working with continuous time models. Firstly, even the simple continuous time stochastic model specifications (like the base-case model used in this paper) do not have known densities readily usable to evaluate the likelihood function. Secondly, one often has to resort to using discretisation schemes to simulate state vector (i.e., particle) paths. However, despite this second hurdle, approximating transition densities using discrete distributions consisting of particle support points provides a tractable workaround which can easily be adapted for most of the typical model specifications considered. The filtering distributions provide estimators for the latent states and their possible distributions. The Monte Carlo procedures involved in the simulation of particles can be tailored to accommodate models with jumps and with more complex state dynamics than featured in the Heston model specification.

In the context of the base model specification previously introduced, we outline the following optimal filtering procedure in the spirit of Johannes et al. (2009), Christoffersen et al. (2010) and Hurn et al. (2015) and introduce the necessary implementation details for the Monte Carlo estimation exercise. First, let us denote the log-likelihood function of the model under the physical probability measure by

$$\log \mathcal{L}_T = \log f(X_0) + \sum_{i=0}^T \log f(X_{i+1}|X_i; \theta^{\mathbb{P}, \mathbb{P} \cap \mathbb{Q}}). \quad (1.15)$$

The evaluation of this likelihood function is unfeasible as the state vector is not fully observed. Also note that it does not provide any information about $\theta^{\mathbb{Q}}$, the parameters which feature in the dynamics of the model under the risk neutral probability measure.

To take into account option prices and to allow for the likelihood to be evaluated, a filtering rule using the time series observations for the components of the state vector which are observable is introduced alongside an option price panel likelihood observation in order to extract information about the latent states which are not observable. Let us denote by $\mathcal{O}_i$ the filtration containing the collection of sigma algebras generated by the time series of observable states and associated option prices. E.g., for our base model specification this is the filtration generated by $S_1, \dots, S_T$ and the panel $p_1(T, K), \dots, p_T(T, K)$ of option prices for all available option maturities T and strike prices K . The level of the stochastic volatility process V_i is unobservable. To evaluate the likelihood contribution, the following need to be evaluated:

Step 1. T_{i+1} likelihood contribution:

$$\begin{aligned} f(X_{i+1}|\mathcal{O}_i; \theta) &= \int_{\mathcal{D}_V} f(S_{i+1}|V_{i+1}, \mathcal{O}_i; \theta) f(V_{i+1}|\mathcal{O}_i; \theta) dV_{i+1} \\ &= \int_{\mathcal{D}_V} f(S_{i+1}|V_{i+1}, \mathcal{O}_i; \theta) \int_{\mathcal{D}_V} f(V_{i+1}|V_i, \mathcal{O}_i; \theta) \\ &\quad \times f(V_i|\mathcal{O}_i; \theta) dV_i dV_{i+1} \end{aligned} \quad (1.16)$$

Step 2. Latent state variable density update:

$$f(V_{i+1}|\mathcal{O}_{i+1}; \theta) = \frac{f(X_{i+1}, V_{i+1}|\mathcal{O}_{i+1}; \theta)}{f(X_{i+1}|\mathcal{O}_{i+1}; \theta)} \quad (1.17)$$

1.3.3.1 Current computational implementation.

At the current stage, the Heston simulation engine, the option-panel generation routines and the first-step implied-state C-GMM estimator have been implemented in Python. The estimator constructs scalar date-by-date implied-variance states from option-implied volatilities, evaluates the analytic Heston $\mathbb{P}$ -transition characteristic function and forms a first-step C-GMM objective with analytically integrated Gaussian instruments. A 32-step Runge–Kutta implementation is used as a numerical check. For each of the 100 samples, the Monte Carlo exercise uses at most three deterministic starting values and a bounded derivative-free two-pass search under the clean panel and the three observation-error designs. The resulting parameter, implied-state and risk-premium accuracy results are reported in the next section.

2 Monte Carlo Results for the Implied-State Estimator

2.1 Monte Carlo design and reporting conventions

The Monte Carlo exercise consists of 100 independent samples and four option-panel designs: the clean benchmark, low i.i.d. observation error, spatially correlated observation error and persistent-factor observation error. Each sample contains 526 weekly state observations and 15 option contracts per date. After constructing the estimation state, the 525 physical return intervals provide 524 conditional residual transitions for the first-step C-GMM criterion. Within each sample, the same simulated path is used for all four designs.

A sample-design estimate is included when its record, simulated path and option panel are complete, the stored files agree with their recorded checksums and all required parameter, implied-state and criterion values are finite. On this basis, all 100 replications are included for each design. The accuracy statistics are first computed within each replication and are then summarised across replications. Throughout the reported accuracy tables, MAE denotes mean absolute error. The parameter table follows the summaries used by Boswijk et al. (2024) and also reports bias and RMSE. The results concern the accuracy of the point estimates; standard errors and confidence intervals are not considered.

2.2 Structural-parameter estimates

Statistic	η	κ	$\bar{V}$	σ_v	ρ	$\kappa_{\mathbb{Q}}$
<i>Panel A: Clean ($n = 100$)</i>						
True value	5.000	7.000	0.0225	0.400	-0.500	2.000
Mean	5.142	7.024	0.0218	0.337	-0.536	1.582
Median	4.964	7.032	0.0215	0.352	-0.550	1.500
Bias	0.142	0.024	-0.0007	-0.063	-0.036	-0.418
Empirical SD	1.725	1.242	0.0025	0.060	0.137	0.139
RMSE	1.722	1.236	0.0026	0.087	0.141	0.440
10% quantile	2.937	5.330	0.0192	0.232	-0.741	1.500
90% quantile	7.444	8.978	0.0254	0.398	-0.353	1.796
<i>Panel B: Design A: low i.i.d. ($n = 100$)</i>						
True value	5.000	7.000	0.0225	0.400	-0.500	2.000
Mean	5.534	6.703	0.0201	0.384	-0.531	1.576
Median	5.319	6.597	0.0198	0.416	-0.524	1.506
Bias	0.534	-0.297	-0.0024	-0.016	-0.031	-0.424
Empirical SD	1.776	1.121	0.0018	0.103	0.139	0.148
RMSE	1.846	1.154	0.0030	0.104	0.141	0.449
10% quantile	3.341	5.235	0.0182	0.230	-0.757	1.500
90% quantile	8.000	8.150	0.0227	0.502	-0.359	1.682
<i>Panel C: Design B: spatial correlation ($n = 100$)</i>						
True value	5.000	7.000	0.0225	0.400	-0.500	2.000
Mean	5.513	6.711	0.0200	0.387	-0.510	1.584
Median	5.294	6.367	0.0197	0.414	-0.505	1.506
Bias	0.513	-0.289	-0.0025	-0.013	-0.010	-0.416
Empirical SD	1.768	1.154	0.0018	0.109	0.141	0.151
RMSE	1.832	1.184	0.0031	0.109	0.141	0.442
10% quantile	3.351	5.389	0.0180	0.235	-0.726	1.500
90% quantile	8.000	8.426	0.0228	0.528	-0.315	1.813
<i>Panel D: Design C: persistent factor ($n = 100$)</i>						
True value	5.000	7.000	0.0225	0.400	-0.500	2.000
Mean	5.495	6.709	0.0200	0.386	-0.527	1.576
Median	5.323	6.581	0.0197	0.417	-0.527	1.523
Bias	0.495	-0.291	-0.0025	-0.014	-0.027	-0.424
Empirical SD	1.773	1.114	0.0019	0.103	0.137	0.129
RMSE	1.832	1.146	0.0031	0.103	0.139	0.443
10% quantile	3.403	5.348	0.0179	0.239	-0.733	1.500
90% quantile	8.000	8.386	0.0226	0.497	-0.336	1.682

Table 1: Distribution of the first-step IS-CGMM parameter estimates across the 100 samples. The risk-neutral parameter is $\kappa_{\mathbb{Q}} = \kappa - \eta_V$. Bias, empirical standard deviation and RMSE are calculated separately for each design.

For the clean panel, the estimates of κ and $\bar{V}$ are close to their true values on average, while the estimates of η show more variation. Relative to the clean benchmark, the three observation-error designs also move the mean estimate of $\bar{V}$ below its true value and increase the RMSE of σ_V .

2.3 Accuracy of implied variance states

For each replication, the final implied variance sequence is matched by date to the simulated variance sequence. Table 2 reports the average across replications of the five accuracy statistics first calculated within each sample. Figure 1 shows the distribution of the corresponding RMSE values across samples.

Design	Mean error	MAE	RMSE	Median abs.	95% quantile
Clean	-0.052	0.173	0.193	0.165	0.306
Design A	-0.264	0.323	0.376	0.306	0.668
Design B	-0.267	0.323	0.376	0.304	0.674
Design C	-0.263	0.321	0.374	0.305	0.669

Table 2: Accuracy of the final implied variance sequence. The error measures are first calculated within each sample using 526 matched dates and are then averaged across the 100 samples. The 95% quantile is calculated from the absolute errors. Variance errors are multiplied by 100.

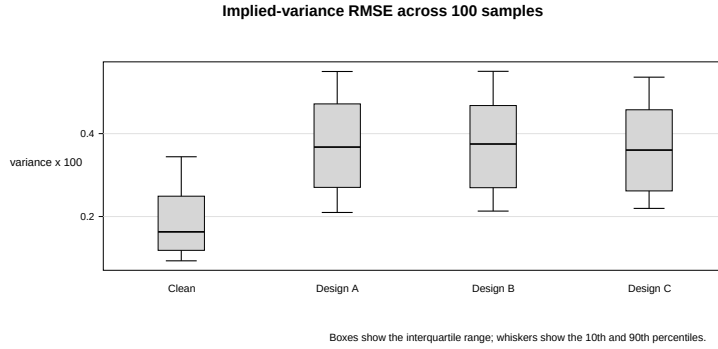


Figure 1: Distribution of implied variance RMSE across the 100 samples. The boxes show the interquartile range, the centre lines show the medians and the whiskers show the 10th and 90th percentiles. Variance errors are multiplied by 100.

The clean benchmark has the smallest average RMSE. Observation error reduces the

accuracy of the implied variance states, but the average errors are similar for the three observation-error designs.

2.4 Accuracy of risk-premium estimates

For each replication, the one-month equity and variance risk-premium paths are computed using the definitions in Appendix C. The transformations are applied separately to the parameter estimates and implied variance path from that replication. The corresponding true path is computed using the simulated variance and the true parameters. An error is defined as the estimated value minus the true value. The variance risk premium follows the physical-minus-risk-neutral convention in Boswijk et al. (2016).

Design	Truth	Estimate	Mean error	MAE	RMSE
<i>Panel A: ERP: annualised percentage points</i>					
Clean	11.0895	10.9129	-0.1765	3.1138	3.5002
Design A	11.0895	10.7033	-0.3861	3.3412	3.9496
Design B	11.0895	10.6397	-0.4497	3.2898	3.8910
Design C	11.0895	10.6221	-0.4673	3.3065	3.9011
<i>Panel B: VRP: annualised variance multiplied by 100</i>					
Clean	-0.4368	-0.4589	-0.0221	0.0696	0.0827
Design A	-0.4368	-0.3950	0.0418	0.0764	0.0932
Design B	-0.4368	-0.3949	0.0420	0.0795	0.0965
Design C	-0.4368	-0.3951	0.0417	0.0758	0.0921

Table 3: Accuracy of the one-month equity and variance risk premia. Each measure is first calculated from the matched estimated and true paths within a sample and is then averaged across the 100 samples. The variance risk premium is defined as physical minus risk-neutral expected average variance.

The average ERP errors are small relative to their variation across replications, although the ERP RMSE remains sizeable. The average VRP error is negative for the clean benchmark and positive for each of the three observation-error designs.

2.5 Summary of results

For the clean benchmark, the implied-state estimator recovers the weekly variance path closely. Under the three observation-error designs, observation error approximately doubles the average RMSE of the implied states and somewhat reduces the accuracy of the parameter and risk-premium estimates.

A Simulation of Heston Paths and Model Option Panels

This appendix documents the simulation procedure used to generate paths of the state vector $X_t = [\log(S_t), V_t]$ under the historical probability measure $\mathbb{P}$ and the associated panel of model-implied European option prices under the equivalent martingale measure $\mathbb{Q}$, based on the data-generating process (1.1)–(1.2) and pricing dynamics (1.3)–(1.4).

A.1 Sampling grid and contract set

Let $\{t_i\}_{i=0}^T$ denote an equally-spaced time grid with $t_{i+1} - t_i = \Delta > 0$. The simulated sample consists of $\{(\log(S_{t_i}), V_{t_i})\}_{i=0}^T$. At each date t_i , we define a (possibly time-varying) set of option contracts $\mathcal{C}_i = \{(T_i^{(j)}, K_i^{(j)}, \text{type}^{(j)})\}_{j=1}^{q_i}$ and compute a vector of model-implied prices $p_i^{\text{model}} \in \mathbb{R}^{q_i}$.

In the numerical design, 100 independent samples are generated. For each sample, the Heston state vector is simulated at a daily frequency and retained at a weekly frequency. The daily simulation step is set to

$$\delta = \frac{1}{252},$$

while weekly observations are obtained by retaining every fifth daily simulated value. The retained sample contains 525 weekly intervals, corresponding to 526 weekly state observations. A burn-in of 756 daily observations is used. The initial spot level is set to $S_0 = 100$, and the daily values of the simulated state vector are stored in addition to the weekly observations.

At each weekly observation date, the clean option panel contains options with maturities:

$$\mathcal{T} \in \left\{ \frac{1}{12}, \frac{1}{4}, \frac{1}{2} \right\}$$

and log-forward moneyness levels:

$$m \in \{-0.15, -0.075, 0, 0.075, 0.15\}.$$

The option type is selected using an out-of-the-money convention. Put options are used for negative log-forward moneyness values and call options are used for non-negative log-forward moneyness values. At-the-money options are treated as calls. Thus, each weekly panel contains 15 option contracts.

A.2 State simulation under $\mathbb{P}$

Let $y_t := \log(S_t)$ and write $y_i := y_{t_i}$ and $V_i := V_{t_i}$. Under $\mathbb{P}$, the dynamics are given by (1.1)–(1.2). We simulate V_{i+1} using the quadratic-exponential scheme of Andersen (2008) and then update y_{i+1} using a discretisation that preserves the leverage channel implied by ρ .

A.2.1 Quadratic-exponential draw for $V_{i+1} \mid V_i$

Let $e_i = \exp(-\kappa\Delta)$. Conditional on V_i , the first two moments of the square-root variance transition are

$$m_i = \bar{V} + (V_i - \bar{V})e_i, \quad (A.1)$$

$$s_i^2 = \frac{V_i \sigma_V^2 e_i (1 - e_i)}{\kappa} + \frac{\bar{V} \sigma_V^2 (1 - e_i)^2}{2\kappa}, \quad \psi_i = \frac{s_i^2}{m_i^2}. \quad (A.2)$$

The production implementation uses the branch threshold $\psi_c = 1.5$. When $\psi_i \leq \psi_c$, define

$$b_i^2 = \frac{2}{\psi_i} - 1 + \sqrt{\frac{2}{\psi_i}} \sqrt{\frac{2}{\psi_i} - 1}, \quad a_i = \frac{m_i}{1 + b_i^2}, \quad (A.3)$$

and draw $V_{i+1} = a_i(\sqrt{b_i^2} + Z_i)^2$ with $Z_i \sim \mathcal{N}(0, 1)$. When $\psi_i > \psi_c$, set

$$p_i = \frac{\psi_i - 1}{\psi_i + 1}, \quad \beta_i = \frac{1 - p_i}{m_i}. \quad (A.4)$$

For $U_i \sim \mathcal{U}(0, 1)$, the draw is zero if $U_i \leq p_i$ and is

$$V_{i+1} = -\frac{1}{\beta_i} \log \left(\frac{U_i - p_i}{1 - p_i} \right) \quad (A.5)$$

otherwise. Both branches preserve non-negativity and match the conditional moments used by the scheme.

A.2.2 Integrated variance proxy

Define the integrated variance over $[t_i, t_{i+1}]$ by

$$I_i := \int_{t_i}^{t_{i+1}} V_s \, ds. \quad (A.6)$$

Since I_i is not sampled explicitly, we use the trapezoidal proxy

$$\widehat{I}_i := \frac{\Delta}{2} (V_i + V_{i+1}). \quad (\text{A.7})$$

A.2.3 Log-price update with leverage

Let $Z_{i+1} \sim \mathcal{N}(0, 1)$ i.i.d. across i and independent of the variance draw. Using the decomposition of the Brownian drivers in (1.1)–(1.2), the log-price increment is simulated as

$$\begin{aligned} \Delta y_i &:= y_{i+1} - y_i \\ &= (r_{t_i} - q_{t_i})\Delta + \left(\eta - \frac{1}{2}\right)\widehat{I}_i + \frac{\rho}{\sigma_V} \left[(V_{i+1} - V_i) - \kappa (\bar{V}\Delta - \widehat{I}_i) \right] \\ &\quad + \sqrt{1 - \rho^2} \sqrt{\widehat{I}_i} Z_{i+1}. \end{aligned} \quad (\text{A.8})$$

Finally, set $y_{i+1} = y_i + \Delta y_i$ and $S_{i+1} = \exp(y_{i+1})$.

A.3 Model option panels under $\mathbb{Q}$

Given the simulated state $X_i = [\log(S_i), V_i]$ at time t_i , model option prices are computed under $\mathbb{Q}$ for each contract $(T_i^{(j)}, K_i^{(j)}) \in \mathcal{C}_i$ by

$$p_i^{\text{model}}(T_i^{(j)}, K_i^{(j)}) = P\left(X_i, T_i^{(j)}, K_i^{(j)}; \theta^{\mathbb{P} \cap \mathbb{Q}}, \theta^{\mathbb{Q}}\right), \quad j = 1, \dots, q_i, \quad (\text{A.9})$$

with $\theta = [\eta, \kappa, \bar{V}, \sigma_V, \rho, \eta_V]$ and the partitioning used in the main text. Under $\mathbb{Q}$, the variance drift (1.4) can be written in standard Heston form by defining

$$\kappa_{\mathbb{Q}} := \kappa - \eta_V, \quad \bar{V}_{\mathbb{Q}} := \frac{\kappa \bar{V}}{\kappa - \eta_V}, \quad (\text{A.10})$$

so that $dV_t = \kappa_{\mathbb{Q}}(\bar{V}_{\mathbb{Q}} - V_t) dt + \sigma_V \sqrt{V_t} dW_{t,v}^{\mathbb{Q}}$.

A.3.1 COS pricing

European option prices in (A.9) are evaluated using the Fourier-cosine (COS) method, which expands the risk-neutral density of the log-price (or log-return) on a bounded interval as a cosine series and integrates the payoff against this approximation.

Let $\mathcal{T} := T - t_i$ and define the log-forward moneyness

$$x := \log\left(\frac{S_i e^{(r_{t_i} - q_{t_i})\mathcal{T}}}{K}\right). \quad (\text{A.11})$$

Denote by $\phi(u; \mathcal{T}, X_i)$ the conditional characteristic function of $\log(S_T)$ under $\mathbb{Q}$ (which is available in semi-closed form for Heston). The COS approximation for a European call can be written as

$$C_i(T, K) \approx e^{-r_{t_i}\mathcal{T}} \sum_{k=0}^{N-1} {}'\Re\left\{\phi\left(\frac{k\pi}{b-a}; \mathcal{T}, X_i\right) \exp\left(-i\frac{k\pi a}{b-a}\right)\right\} U_k(x), \quad (\text{A.12})$$

where $[a, b]$ is a truncation range for the log-price (or log-return) support, N is the number of cosine terms, and $\sum'$ indicates that the $k = 0$ term is weighted by $1/2$. The payoff coefficients $U_k(x)$ admit closed forms for vanilla calls and puts. Put prices are obtained either via the corresponding COS payoff coefficients or via put-call parity.

A.4 Parameter values used for clean sample generation

The historical-measure parameter values used in the data generating process are

$$\eta = 5.0, \quad \kappa = 7.0, \quad \bar{V} = 0.0225, \quad \sigma_V = 0.4, \quad \rho = -0.5,$$

with ($r=0.02$) and ($q=0$). The risk-neutral variance risk-premium parameter is set to

$$\eta_V = 5.0.$$

Given the notation in (1.4), this implies

$$\kappa_{\mathbb{Q}} = \kappa - \eta_V = 2.0, \quad \bar{V}_{\mathbb{Q}} = \frac{\kappa \bar{V}}{\kappa - \eta_V} = 0.07875.$$

Clean option prices are computed using the COS method. The corresponding Black-Scholes implied volatilities are computed using the same implied-volatility inversion routine used for noisy panels; see (Jäckel, 2015).

B Option Panel Observation Error Specifications

This appendix details the functional forms used to generate noisy option panels from the clean model-implied option panel. The purpose is to introduce observation error

structures which are simple enough to be implemented in a large-scale Monte Carlo design, but sufficiently close to observed option-market features to provide a meaningful stress-test of PF and IS estimation.

Noisy option panels are constructed from the clean model-implied panels generated in Appendix A using the observation error designs introduced in the main text. The clean panel at date t_i consists of model prices p_i^0 , Black-Scholes implied volatilities σ_i^0 , contract characteristics and Black-Scholes vegas. The observation-error functional is applied to the clean implied-volatility value before the corresponding raw noisy price is obtained from the Black-Scholes pricing formula. No-arbitrage projection is then applied to the price when required, followed by implied-volatility inversion to recover the observed noisy implied volatility, using the same procedure for all four designs.

B.1 Price validity and implied-volatility inversion

Let $L_i^{(j)}$ and $U_i^{(j)}$ denote the no-arbitrage lower and upper bounds for the corresponding European option. For a call option,

$$L_i^{(j)} = \max \left(S_i e^{-qT_i^{(j)}} - K_i^{(j)} e^{-rT_i^{(j)}}, 0 \right), \quad U_i^{(j)} = S_i e^{-qT_i^{(j)}}.$$

For a put option,

$$L_i^{(j)} = \max \left(K_i^{(j)} e^{-rT_i^{(j)}} - S_i e^{-qT_i^{(j)}}, 0 \right), \quad U_i^{(j)} = K_i^{(j)} e^{-rT_i^{(j)}}.$$

The observed price is obtained by projecting the raw noisy price into the admissible interval:

$$p_i^{(j),\text{obs}} = \min \left\{ U_i^{(j)} - \epsilon_p, \max \left[L_i^{(j)} + \epsilon_p, \tilde{p}_i^{(j)} \right] \right\}. \quad (\text{B.1})$$

The projection in (B.1) is used instead of the theoretically possible alternative of re-sampling the random quantities in the observation-error mechanism. This keeps the distributional draws reproducible and preserves information about the frequency with which the noise mechanism generates boundary prices. The implementation therefore records whether capping is applied at the lower or upper bound.

The observed Black-Scholes implied volatility $\sigma_i^{(j),\text{obs}}$ is computed from $p_i^{(j),\text{obs}}$ using a robust implied-volatility inversion routine. In the implementation this is handled through a stand-alone implied-volatility solver interface so that the same routine can

be used for clean panels, noisy panels and later estimator diagnostics. The preferred numerical method is the rational approximation approach of Jäckel (2015), which is designed for fast and robust implied-volatility inversion.

B.2 Implementation of noisy option panels

The implementation records, for each contract, the clean model price, the clean Black-Scholes implied volatility, the raw noisy price before validity enforcement, the capped observed price, the observed implied volatility, the noise scenario and indicators for whether lower- or upper-bound capping was applied.

The same implied-volatility inversion routine is used for clean and noisy panels. This is important because the subsequent IS and PF estimation exercises may use observed implied volatilities, price-space errors, or vega-scaled price errors. A single stand-alone implied-volatility solver therefore prevents the construction of different data sets from depending on different numerical inversion conventions.

B.3 Parameter Choices for the Observation Error Specifications

For Design A, the scale parameters are set to

$$\alpha_0 = 0.0015, \quad \alpha_m = 2.0, \quad \alpha_{\mathcal{T}} = 0.05.$$

The value of α_0 fixes the baseline magnitude of the implied-volatility observation error. Expressed in decimal volatility units, $\alpha_0 = 0.0015$ corresponds to 15 implied-volatility basis points before the moneyness and maturity adjustments are applied. The parameters α_m and $\alpha_{\mathcal{T}}$ determine the relative variation of the error scale across the option panel. Given the log-moneyness grid used in the simulation design, $\alpha_m = 2.0$ implies that a contract with moneyness $|m| = 0.15$ has an implied-volatility observation error scale which is approximately 30 percent larger than that of an otherwise identical at-the-money contract. The maturity parameter $\alpha_{\mathcal{T}} = 0.05$ implies a smaller adjustment which is largest for short-dated contracts. The maturity component is approximately 17 percent at $\mathcal{T} = 1/12$, 10 percent at $\mathcal{T} = 1/4$ and 7 percent at $\mathcal{T} = 1/2$.

For Design B, the marginal scale parameters α_0 , α_m and $\alpha_{\mathcal{T}}$ are kept equal to those used for Design A. The difference relative to Design A is therefore introduced through the dependence structure of the observation error, rather than through a different marginal

error scale. The correlation length parameters are set to

$$\ell_m = 0.15, \quad \ell_{\mathcal{T}} = \frac{1}{3}.$$

Given the log-moneyness grid used in the simulation design, this implies that two adjacent moneyness points with the same maturity have an observation error correlation of approximately 0.61. The maturity length parameter implies correlations of approximately 0.61 between the one-month and three-month maturities, 0.47 between the three-month and six-month maturities, and 0.29 between the one-month and six-month maturities. The resulting correlation matrix therefore induces local dependence across the option panel, while avoiding an almost perfectly correlated surface error.

For Design C, the persistence matrix is set to

$$A = \text{diag}(0.85, 0.65, 0.65).$$

The first diagonal element governs the persistence of the common level component of the observation error, while the second and third diagonal elements govern the persistence of the moneyness-slope and maturity-slope components. The level component is made more persistent than the slope components, reflecting the interpretation of this design as one in which part of the option panel observation error is carried across adjacent observation dates.

The innovation covariance matrix Q is specified through the stationary standard deviations of the three factor components. These are set to

$$(\bar{s}_{f,0}, \bar{s}_{f,m}, \bar{s}_{f,\mathcal{T}}) = (0.0007, 0.0025, 0.0008).$$

Given $A = \text{diag}(a_0, a_m, a_{\mathcal{T}})$, the innovation covariance matrix is

$$Q = \text{diag}\left((1 - a_0^2)\bar{s}_{f,0}^2, (1 - a_m^2)\bar{s}_{f,m}^2, (1 - a_{\mathcal{T}}^2)\bar{s}_{f,\mathcal{T}}^2\right).$$

This gives

$$Q = \text{diag}\left(1.35975 \cdot 10^{-7}, 3.609375 \cdot 10^{-6}, 3.696 \cdot 10^{-7}\right).$$

The residual idiosyncratic scale is chosen so that Design C has the same unconditional pointwise error scale as Design A. Let

$$\Omega_f = \text{diag}\left(\bar{s}_{f,0}^2, \bar{s}_{f,m}^2, \bar{s}_{f,\mathcal{T}}^2\right).$$

Then

$$s_{\sigma,u,i}^{(j)} = \sqrt{\max \left[\left(s_{\sigma,i}^{(j)} \right)^2 - \left(b_i^{(j)} \right)^\top \Omega_f b_i^{(j)}, 0 \right]}.$$

Thus, the common component introduces temporal dependence and common variation across moneyness and maturity, while the total marginal implied-volatility observation error remains comparable to that of Designs A and B. This makes the comparison across the four designs depend mainly on the dependence structure of the observation error rather than on a mechanical increase in its absolute scale.

For Design D, the parameters governing the persistent-factor specification are kept equal to those used for Design C. The additional variance-linked component is specified as

$$f_{V,i} = \gamma_V z_{V,i},$$

where $z_{V,i}$ is the standardised latent variance state. The parameter γ_V controls the magnitude of the state-dependent component. Its numerical value is chosen to generate mild dependence between option observation errors and the latent variance state. The idiosyncratic residual variance is reduced accordingly so that the unconditional pointwise error scale remains comparable to that of Designs A–C.

B.4 Illustration of the Observation Error Specifications

This subsection reports results based on all 100 samples. Signed implied-volatility error is defined as observed minus model implied volatility. The surface results retain the common moneyness and maturity points. The two dependence measures in Figure 2 are calculated separately within each sample, so the boxes show how they vary across the 100 samples.

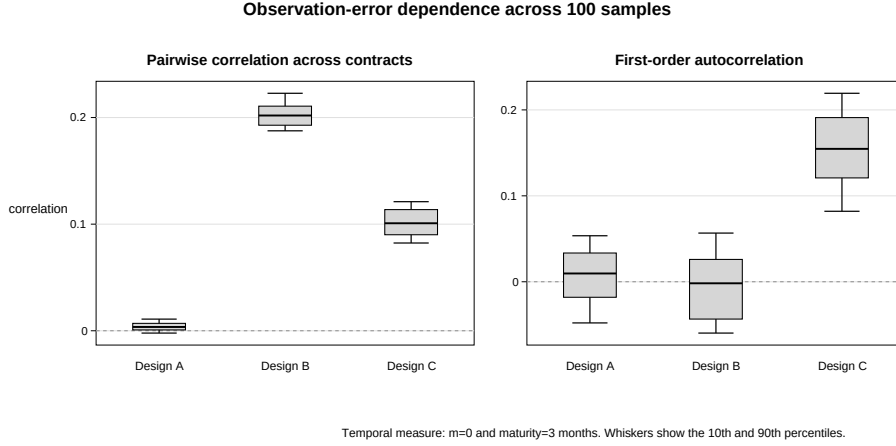


Figure 2: Dependence in the observation errors across the 100 samples. For each sample, the left panel averages the pairwise correlations between the error series for the 15 contracts. The right panel reports the first-order autocorrelation of the error series for the at-the-money three-month contract. Each box summarises the resulting sample-level values; the centre line is the median and the whiskers show the 10th and 90th percentiles.

Design	Mean	MAE	RMSE	Median abs.	95% quantile	Maximum abs.
Clean	0.00	0.00	0.00	0.00	0.00	0.00
Design A	-100.52	116.63	377.08	14.74	1232.96	1907.80
Design B	-100.63	116.69	377.31	14.70	1233.55	1926.17
Design C	-100.64	116.67	377.21	14.72	1233.20	1907.94

Table 4: Implied-volatility observation errors across the 100 samples, measured in implied-volatility basis points. Signed error is observed minus model implied volatility, and the 95% quantile is calculated from the absolute errors. For each design, the statistics combine the errors from all 100 samples, 526 dates and 15 contracts, giving 789,000 observations. The clean benchmark is zero by construction.

C Risk-Premium Definitions

This appendix gives the definitions of the one-month equity and variance risk premia used in Section 2. They follow the pure-diffusion component in Boswijk et al. (2016). For a horizon $h = T - t$, write the expected average variance under $\mathbb{P}$, conditional on

$V_t = v$, as

$$\text{AV}^{\mathbb{P}}(v, h) = \frac{1}{h} \mathbb{E}^{\mathbb{P}} \left[\int_t^T V_s \, ds \middle| V_t = v \right] = \bar{V} + (v - \bar{V}) \frac{1 - \exp(-\kappa h)}{\kappa h}. \quad (\text{C.1})$$

Under $\mathbb{Q}$, define

$$\kappa_{\mathbb{Q}} = \kappa - \eta_V, \quad \bar{V}_{\mathbb{Q}} = \frac{\kappa \bar{V}}{\kappa_{\mathbb{Q}}}. \quad (\text{C.2})$$

The corresponding conditional expected average variance is

$$\text{AV}^{\mathbb{Q}}(v, h) = \bar{V}_{\mathbb{Q}} + (v - \bar{V}_{\mathbb{Q}}) \frac{1 - \exp(-\kappa_{\mathbb{Q}} h)}{\kappa_{\mathbb{Q}} h}. \quad (\text{C.3})$$

For the Heston specification in Section 1, the horizon- h equity risk premium and the physical-minus-risk-neutral variance risk premium are therefore

$$\text{ERP}_t(h) = \eta \text{AV}^{\mathbb{P}}(V_t, h), \quad (\text{C.4})$$

$$\text{VRP}_t(h) = \text{AV}^{\mathbb{P}}(V_t, h) - \text{AV}^{\mathbb{Q}}(V_t, h). \quad (\text{C.5})$$

The reported horizon is $h = 1/12$. ERP is annualised and multiplied by 100 to obtain percentage points. VRP is annualised variance and is also multiplied by 100. For each replication, the estimated paths are calculated from its parameter estimate and final implied state sequence, while the true paths are calculated from its simulated variance sequence and the true parameter vector. The transformations in (C.1)–(C.5) are therefore carried out before the results are summarised across replications.

D Implementation of the IS C-GMM Estimator

This appendix records the numerical implementation of the implied-state C-GMM estimator introduced in Section 1.3.2. The purpose is to separate the computational choices needed for the Monte Carlo exercise from the main description of the estimator.

D.1 Construction of implied variance states

For each observation date t_i , the implied-state step solves the scalar minimisation problem in equation (1.13). The implementation uses the observed Black-Scholes implied-volatility panel as the target object. For each candidate variance value v , Heston option

prices are computed under $\mathbb{Q}$ for all contracts in $\mathcal{C}_i$, using the risk-neutral parameter subset $\theta_{\mathbb{Q}}$. These model prices are then inverted into Black-Scholes implied volatilities using the same implied-volatility inversion convention as that used in the construction of the clean and noisy panels. The criterion function is the sum of squared differences between observed and model-implied volatilities across the contracts in the panel for date i .

The minimisation is one dimensional because the Heston design contains a single latent variance state. The search is carried out over a bounded interval $\mathcal{V} = [\underline{V}, \bar{V}]$. Consecutive option panels are close in time, so the numerical search can be "warm-started" from the previously implied variance level. In the clean-panel case, evaluating the state-implying step at the true risk-neutral parameter vector provides a direct numerical check on the implementation, since the recovered sequence should be numerically indistinguishable from the simulated variance path.

The same procedure is applied to clean and noisy option panels. Hence, differences between the implied states recovered from the different observation error designs are driven by the observation error mechanisms themselves and not by changes in the pricing or implied-volatility inversion convention.

D.2 Fixed COS basis

For an option maturity $\mathcal{T}$, the COS frequencies are fixed as

$$u_k(\mathcal{T}) = \frac{k\pi}{2L_{\mathcal{T}}}, \quad k = 0, \dots, N-1, \quad (\text{D.1})$$

where $[-L_{\mathcal{T}}, L_{\mathcal{T}}]$ is the effective interval for log-forward moneyness. The interval does not depend on the candidate variance value. The payoff expansion and the affine coefficients $A_k(\mathcal{T})$ and $B_k(\mathcal{T})$ are therefore prepared once for each maturity and risk-neutral parameter vector and reused across strikes and dates.

The variance domain is $V \in [10^{-6}, 0.15]$. The parameter ranges are $\kappa_{\mathbb{Q}} \in [1.5, 4.0]$, $\bar{V}_{\mathbb{Q}} \in [0.012, 0.24]$, $\sigma_V \in [0.20, 0.65]$ and $\rho \in [-0.80, -0.20]$. The calibration uses a seeded Latin-hypercube design together with the true DGP, the initial parameter value and two moderate perturbations. A Carr–Madan Fourier inversion is used to validate an 8192-term COS reference. Table 5 reports the selected specification. Prices with Black-Scholes vega below 5×10^{-6} are reported separately because numerical price differences close to 10^{-10} produce unstable implied-volatility differences. There are no invalid prices

or implied-volatility inversions in the calibration design.

$\mathcal{T}$	$L_{\mathcal{T}}$	N	Max. IV error	Max. IV error, vega screen
1/12	0.75	576	2.23×10^{-4}	1.81×10^{-6}
1/4	1.25	576	2.82×10^{-7}	2.82×10^{-7}
1/2	2.00	576	2.77×10^{-8}	2.77×10^{-8}

Table 5: Selected fixed COS basis. The generation and estimation term counts are both 576. The selection uses the lowest term count and then the narrowest interval satisfying the zero-failure error conditions.

D.3 Heston transition characteristic function under $\mathbb{P}$

The C-GMM moment conditions require the conditional characteristic function of the next estimation state

$$X_{i+1}^{\theta_{\mathbb{Q}}} = \left(R_{i+1}, \widehat{V}_{i+1}^{\theta_{\mathbb{Q}}} \right)^{\top}$$

given $X_i^{\theta_{\mathbb{Q}}}$. In the Heston implementation this is evaluated using the physical-measure dynamics of the return and variance pair. Conditional on the current implied variance $\widehat{V}_i^{\theta_{\mathbb{Q}}}$, the relevant characteristic function is written in affine form as

$$\varphi \left(s, X_i^{\theta_{\mathbb{Q}}}, \Delta; \theta \right) = \exp \left(A(\Delta; s, \theta) + B(\Delta; s, \theta) \widehat{V}_i^{\theta_{\mathbb{Q}}} \right),$$

where $s = (s_R, s_V)^{\top}$. The affine coefficients solve the Riccati system associated with the Heston dynamics under $\mathbb{P}$. With terminal loading $B(0) = is_V$, the scalar Riccati equation for B can be written as

$$\frac{dB}{dt} = a + bB + cB^2,$$

where

$$a = is_R \left(\eta - \frac{1}{2} \right) - \frac{1}{2} s_R^2, \quad b = -\kappa + \rho \sigma_V is_R, \quad c = \frac{1}{2} \sigma_V^2.$$

The corresponding A coefficient satisfies

$$\frac{dA}{dt} = is_R(r - q) + \kappa \bar{V} B.$$

The implementation evaluates this system using the analytic Heston Riccati solution.

D.4 Evaluation of the continuum GMM criterion

Let $\tau = (u, s)$, where $u \in \mathbb{R}^2$ denotes the instrument argument and $s \in \mathbb{R}^2$ denotes the conditional characteristic function argument. The first step criterion in (1.3.2) is evaluated using an integrating density $\pi(\tau)$. In the numerical implementation this density is chosen so that

$$\pi(\tau) = \pi_u(u)\pi_s(s),$$

with Gaussian weights for the instrument component. The integration over the instrument argument can then be carried out analytically. Specifically, the products of instrument functions generate terms of the form

$$\int \exp\left(iu^\top \left(X_i^{\theta_Q} - X_j^{\theta_Q}\right)\right) \pi_u(u) \, du,$$

which reduce to a Gaussian kernel in the distance between the two implied-state observations. This leaves a finite dimensional numerical integration over the s -argument of the conditional characteristic function. The production implementation uses two-dimensional Gauss–Hermite quadrature of order 4 with scale vector (1.0, 4.0) for this remaining integral. Integration over the instrument component remains analytic.

This construction follows the same principle as in the continuum-GMM implementation of Boswijk et al. (2016): the criterion uses the conditional characteristic function to generate a continuum of moment restrictions, while numerical integration and analytic simplifications are used to keep the criterion evaluation computationally feasible.

D.5 Parameter transformation and numerical minimisation

The first-step estimator minimises $Q_T^{(1)}(\theta)$ over the admissible parameter set Θ . In the numerical implementation, the constrained Heston parameters are mapped to an unconstrained or bounded free vector. The transformation is based on

$$(\eta, \log \kappa, \log \bar{V}, \log \sigma_V, \operatorname{atanh} \rho, \log \kappa_Q),$$

where

$$\kappa_Q = \kappa - \eta_V, \quad \eta_V = \kappa - \kappa_Q.$$

This parameterisation enforces positivity of the mean-reversion, long-run variance, vol-of-vol and risk-neutral mean-reversion parameters, while keeping ρ inside $(-1, 1)$. The criterion is deterministic for a fixed simulated panel, but it is expensive because each

function evaluation requires a full reconstruction of the implied variance sequence. The implementation uses at most three deterministic starts. It then applies two bounded derivative-free passes and records objective values, implied-state diagnostics and timing information.

At the present stage the Monte Carlo design uses the first-step C-GMM criterion. Efficient second-step weighting is conceptually standard in the continuum-GMM setting, but is deferred in the current implementation in order to keep the comparison with the particle-filter estimator focused on the finite-sample consequences of the state recovery step and the option-panel observation error.

D.6 Production computing arrangement

The completed Monte Carlo exercise consists of 100 independent samples, four panel designs and 400 estimation tasks. It was carried out on an Amazon EC2 `c7i.8xlarge` instance with 32 vCPUs and 64 GiB of memory. Twenty-eight samples were processed in parallel. Each process handled one sample at a time, while each numerical library was restricted to one thread. The calculation could be resumed separately by sample and scenario. Persistent EBS storage was used during the calculation and completed results were copied to private S3 storage. These computing arrangements do not affect the numerical specification, which is the same for every task included in the results.

References

- Ait-Sahalia, Y. and Kimmel, R. (2007). Maximum likelihood estimation of stochastic volatility models. *Journal of financial economics*, 83(2):413–452.
- Andersen, L. B. G. (2008). Simple and efficient simulation of the heston stochastic volatility model. *Journal of Computational Finance*, 11(3):1–42.
- Andersen, T., Fusari, N., Todorov, V., and Varneskov, R. (2021). Spatial dependence in option observation errors. *Econometric Theory*, 37(2):205–247.
- Avellaneda, M., Healy, B., Papanicolaou, A., and Papanicolaou, G. (2020). Pca for implied volatility surfaces.
- Bardgett, C., Gourier, E., and Leippold, M. (2014). Inferring volatility dynamics and risk premia from the s&p 500 and vix markets.
- Bates, D. S. (2006). Maximum likelihood estimation of latent affine processes. *The Review of Financial Studies*, 19(3):909–965.
- Boswijk, H. P., Laeven, R. J. A., and Lalu, A. (2016). Asset returns with self-exciting jumps: Option pricing and estimation with a continuum of moments. *University of Amsterdam mimeo*.
- Boswijk, H. P., Laeven, R. J. A., and Vladimirov, E. (2024). Estimating option pricing models using a characteristic function-based linear state space representation. *Journal of Econometrics*, 244(1):105864.
- Chong, C. H. and Todorov, V. (2024). Volatility of volatility and leverage effect from options. *Journal of Econometrics*, 240(1):105669.
- Christoffersen, P., Jacobs, K., and Mimouni, K. (2010). Volatility dynamics for the s&p500: evidence from realized volatility, daily returns, and option prices. *The Review of Financial Studies*, 23(8):3141–3189.
- Cont, R. and da Fonseca, J. (2002). Dynamics of implied volatility surfaces. *Quantitative Finance*, 2(1):45–60.
- Duarte, J., Jones, C. S., and Wang, J. L. (2024). Very noisy option prices and inference regarding the volatility risk premium. *The Journal of Finance*, 79(5):3581–3621.
- Garcia, R., Lewis, M.-A., Pastorello, S., and Renault, É. (2011). Estimation of objective

- and risk-neutral distributions based on moments of integrated volatility. *Journal of Econometrics*, 160(1):22–32.
- Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2):327–343.
- Hurn, A. S., Lindsay, K. A., and McClelland, A. J. (2015). Estimating the parameters of stochastic volatility models using option price data. *Journal of Business & Economic Statistics*, 33(4):579–594.
- Jäckel, P. (2015). Let’s be rational. *Wilmott*, 2015(75):40–53.
- Johannes, M. S., Polson, N. G., and Stroud, J. R. (2009). Optimal filtering of jump diffusions: Extracting latent states from asset prices. *The Review of Financial Studies*, 22(7):2759–2799.
- Pan, J. (2002). The jump-risk premia implicit in options: Evidence from an integrated time-series study. *Journal of financial economics*, 63(1):3–50.
- Protter, P. E. (2005). *Stochastic integration and differential equations*. Springer.
- Sundaresan, S. M. (2000). Continuous-time methods in finance: A review and an assessment. *The Journal of Finance*, 55(4):1569–1622.